Supervyse OPF IPMI Tool User Guide

Revision 1.0

01/2025

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Revision History

Revision Number	Description	Author	Release Date
1.0	Initial release.	Wade Wu	01, 2025

1 Introduction

This document is intended to guide using Supervyse OPF Tool – **SPV_IPMI**.

IPMI(Intelligent Platform Management Interface) is an open-standard hardware management interface specification that defines a specific way for embedded management subsystems to communicate. IPMI information is exchanged through **BMC**s(Baseboard Management Controllers).

ipmitool is a utility that monitors, configures, and manages devices that support the IPMI. It is an open-source tool for controlling IPMI-enabled systems.

SPV_IPMI is the Insyde Supervyse ipmitool, which is based on ipmitool. It supports a variety of operating systems and can communicate with BMC. In addition, it also enhances the Insyde OEM(Original Equipment Manufacturer) command using the designing 'isc' command to help convert the response data into highly readable text.

1.1 Overview

This document introduces how to use the Insyde customized ipmitool - SPV_IPMI to communicate with Supervyse OPF BMC more easily and effectively.

2 Insyde SPV IPMI User Guide

2.1 Download

Download SPV_IPMI through the Inquire2 tool provided by Insyde. **InQuire2** is a program that provides easy, centralized access to all of Insyde Software's online cloud services.Currently, supported operating systems are Linux, Windows, FreeBSD, DOS, and UEFI.

Take Linux as an Example. There are many tools in the Download service, following the path Tool > ipmitool > **<version>** > OS.

```
I.e. Tool > ipmitool > v1.8.18.24301 > linux > x64.zip
```

At first, click Download.

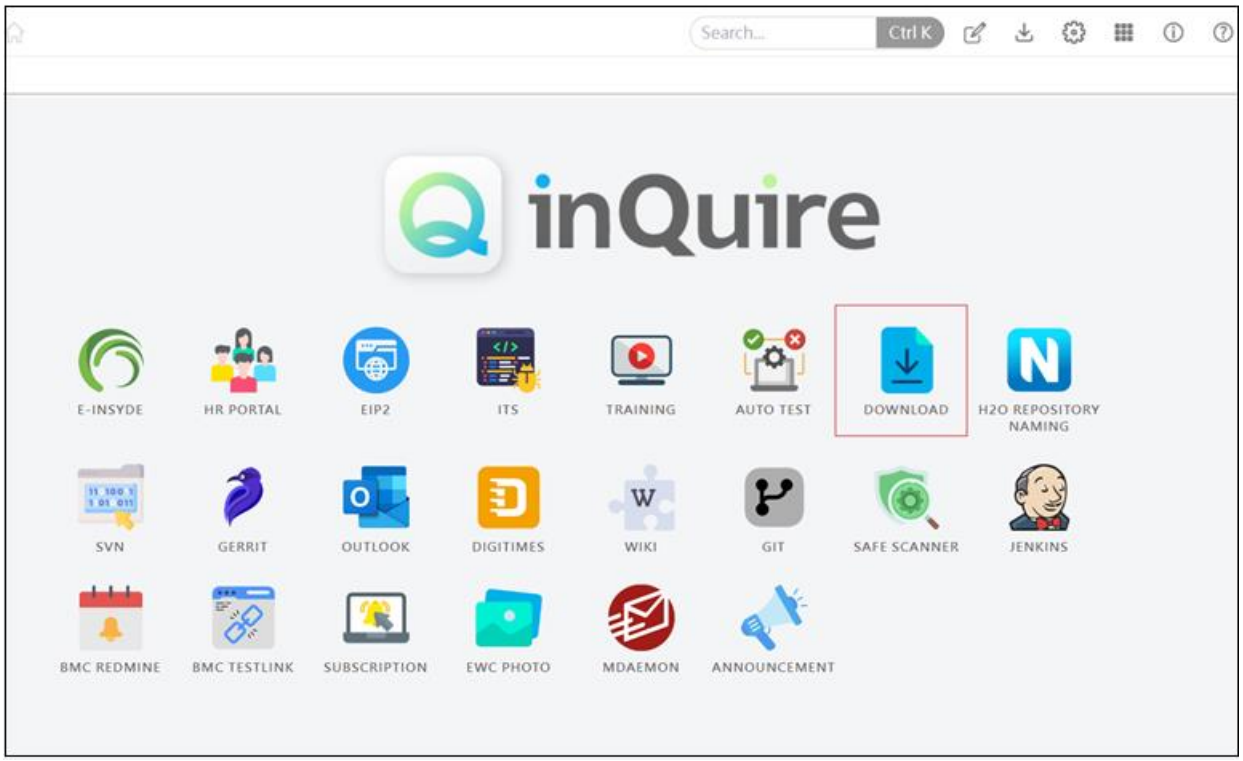

Figure 1-Click Download from InQuire

Then go to Tool > ipmitool > Linux (OS), and click the download button on the right screen.

Unzip it and get the execution file.

If the operating system is Windows, download it using the following path:

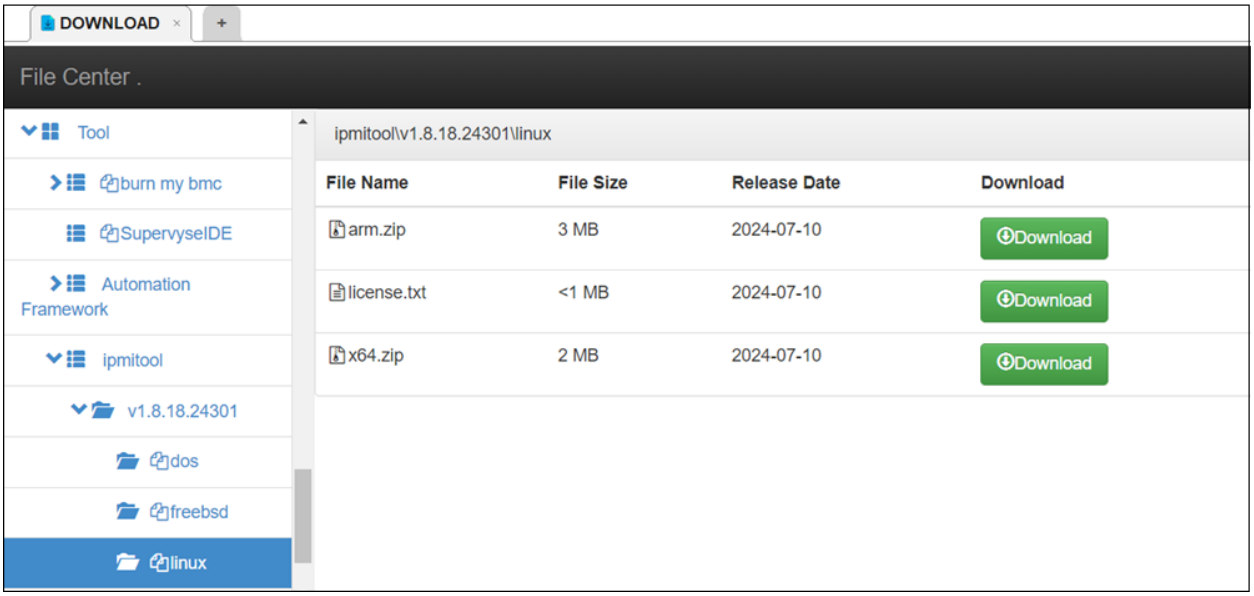

Figure 2-Download from InQuire

```
I.e. Tool > ipmitool > v1.8.18.24301 > Win > x64.zip
```

2.2 Basic Usage

The Insyde customized ipmitool SPV_IPMI usage is very similar to ipmitool.The only difference is extra Insyde utilities are provided by the "isc" command.

"**isc**" is the entry point of Insyde OEM commands.

2.2.1 Execution File

SPV_IPMI provides a variety of execution programs in different operating systems.
The extension of the program running on different operating systems will be different.

Table 1-Executable for different operating system

	Windows	Linux (X64/ARM)	FreeBSD	DOS	UEFI (x64/ARM)
File Name	spv_ipmi.exe	spv_ipmi	spv_ipmi	spv_ipmi.exe	spv_ipmi.efi

NOTE:
1. Take **Linux** and **FreeBSD** as an example. Please confirm that the user has permission to execute on Linux or FreeBSD.

```
# Make the user have execute permission.
# chmod - change file modes.
#   Parameter:
#         x - Set execute bits to enable.
```

```
chmod +x spv_ipmi
```

2. A dot slash is a dot followed immediately by a forward slash (./).
It is used in Linux or FreeBSD to execute a program in the current directory.

```
# Execute a program and show help.
./spv_ipmi -h
```

3. Modify the .bashrc file set environment variable 'PATH' and re-login will executable file without entering './'.

```
# .bashrc file
gedit ~/.bashrc

# Add a line of command to the file '.bashrc' set environment variables.
export PATH=$PATH:./
```

2.2.2 Example

The following examples are basic usages:

```
# Show help
./spv_ipmi -h
```

Please replace the executable file **spv_ipmi** belonging to the current operating system.

There will be different input methods in different operating systems.The command prompt displayed will also be different.

Windows - If the executable file is on the C drive.

```
# Show help
C:\>spv_ipmi.exe -h
```

DOS - If booting via USB and the file is in the root folder.

```
# Show help
C:\>spv_ipmi.exe -h
```

Linux - If the file is in the home directory.

```
# Show help
cd ~
./spv_ipmi -h
```

FreeBSD - If the file is in the home directory.

```
# Show help
```

```
cd ~
./spv_ipmi -h
```

UEFI boot to USB file system(fs0) under UEFI shell.

```
# Show help
fs0:\>spv_ipmi.efi -h
```

2.2.3 Customized

The default Insyde OEM net function code(NetFn) is 0x3E, and the Insyde OEM command code is 0xC0.

Adding a .ini file can change the default value. The Example is listed as follows:

```
; File Name: OEM.ini

[Insyde_OEM_Command]

IPMI_NETFN_ISC_OEM = 0x76
```

The section '[Insyde_OEM_Command]' in the INI file is required. Put the file "OEM.ini" in the SPV_IPMI directory and run SPV_IPMI. It will show the message "NOTE: Update Insyde OEM NetFn from 0x3E to 0x75".

The following table is the keywords of the .ini file:

Table 2-Configuration Keyword for IPMI tool

Key	Description	Default Value
IPMI_NETFN_ISC_OEM	Insyde NetFn	0x3E
IPMI_ISC_OEM_CMD	Insyde Sub-Command ID	0xC0
NETFN_INTEL_OEM_PLATFORM	Intel platform OEM NetFn	0x30
NETFN_CUST_INS	Insyde customer net function	0x38

2.3 Interfaces

SPV_IPMI tool-supported interfaces are visible in the usage help output.

LAN and LAN plus are solutions for remote communication (Out-of-Band), and the others are local solutions (In-Band).

Commonly used interfaces are listed here. For details, please refer to ipmitool README or IPMI SPEC v2.0.

2.3.1 Support

Following is a table of supported interfaces and OS:

Table 3-Supported interfaces and OS

Interface Driver Name	Windows	Linux (X64/ARM)	FreeBSD	DOS	UEFI (X64/ARM)
lan	v	v	v		
lanplus	v	v	v		
ms	v				
imb	v	v			
open		v	v		
kcs				v	v
h2o					v

2.3.2 Description & Usage

2.3.2.1 lan

The lan interface communicates with the BMC over an Ethernet LAN connection, which uses UDP(User Datagram Protocol) under IPv4 or IPv6.

UDP datagrams are formatted to contain IPMI request/response messages with IPMI session headers and RMCP(Remote Management Control Protocol) headers.

The following table is supported parameters:

Table 4-Supported Parameters for lan interface

Parameter	Description	Optional
-I	Specifies the lan communication interface, which supports IPMI v1.5	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device	
-p	The default port of RMCP is 623, and parameter –p can use other ports for connection	V

The following is an example of usage:

```
./spv_ipmi -I lan -H 192.168.9.172 -U root -P 0penBmc
```

2.3.2.2 lanplus

Like the lan interface, the lanplus interface communicates with the BMC over an Ethernet LAN connection using UDP under IPv4 or IPv6. The difference is that the lanplus interface uses the **RMCP+** protocol as described in the IPMI v2.0 specification. RMCP+ allows for improved authentication and data integrity checks, as well as encryption and the ability to carry multiple types of payloads.

Due to the fact that SPV_IPMIbased on ipmitool v1.8.18, adjustments need to be made to the cipher suite in order to communicate with OpenBMC via the network.

- In OpenBmc, the default value for the cipher suite is 17.

The following table shows the available Cipher Suite IDs:

Table 5-Available Cipher Suite IDs for lanplus interface

ID	characteristics	Cipher Suite	Authentication Algorithm	Integrity Algorithm(s)	Confidentiality Algorithm(s)
0	"no password"	00h, 00h, 00h	RAKP-none	None	None
1	S	01h, 00h, 00h	RAKP-HMAC-SHA1	None	None
2	S, A	01h, 01h, 00h		HMAC-SHA1-96	None
3	S, A, E	01h, 01h, 01h			AES-CBC-128
4	S, A, E	01h, 01h, 02h			xRC4-128
5	S, A, E	01h, 01h, 03h			xRC4-40
6	S	02h, 00h, 00h	RAKP-HMAC-MD5	None	None
7	S, A	02h, 02h, 00h		HMAC-MD5-128	None
8	S, A, E	02h, 02h, 01h			AES-CBC-128
9	S, A, E	02h, 02h, 02h			xRC4-128
10	S, A, E	02h, 02h, 03h			xRC4-40
11	S, A	02h, 03h, 00h		MD5-128	None
12	S, A, E	02h, 03h, 01h			AES-CBC-128
13	S, A, E	02h, 03h, 02h			xRC4-128
14	S, A, E	02h, 03h, 03h			xRC4-40
15	S	03h, 00h, 00h	RAKP-HMAC-SHA256	None	None
16	S, A	03h, 04h, 00h		HMAC-SHA256-128	None
17	S, A, E	03h, 04h, 01h			AES-CBC-128
18	S, A, E	03h, 04h, 02h			xRC4-128
19	S, A, E	03h, 04h, 03h			xRC4-40
80h-BFh	OEM specified	OEM specified	OEM specified	OEM specified	OEM specified
C0h-FFh	reserved	-	-	-	-

Key:
S = authenticated session setup (correct role, username and password/key required to establish session)
A = authenticated payload data supported.
E = authentication and encrypted payload data supported

The following table is supported parameters:

Table 6-Supported parameters for lanplus interface

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	

Parameter	Description	Optional
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device	
-p	The default port of RMCP is 623, and parameter –p can use other ports for connection	V
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17
```

2.3.2.3 ms

The interface 'ms' adopts Microsoft Windows built-in driver **Microsoft IPMI driver** (ipmidrv.sys). The Microsoft IPMI driver and WMI(Windows Management Instrumentation) IPMI provider can manipulate BMC from the operating system. For more details about WMI IPMI provider, please see https://docs.microsoft.com/en-us/previous-versions/windows/desktop/ipmiprv/ipmi-provider.

The IPMI driver is a WDM(Windows Driver Model) kernel-mode driver that communicates with the BMC using KCS(Keyboard Controller Style)

NOTE:
1. **Only support MS Windows**.
2. The driver will be automatically generated when the platform supports IPMI.

The following is an example of usage:

```
C:\>spv_ipmi.exe -I ms
```

2.3.2.4 imb

The imb interface supports the Intel IMB (Intel Inter-module Bus) Interface through the /dev/imb device.

- In Windows, The Intel IMB interface is a KCS(Keyboard Controller Style) interface.

The following is an example of usage:

```
C:\>spv_ipmi.exe -I imb
```

2.3.2.5 kcs

The KCS interface works like the communication between a CPU and a keyboard controller in traditional systems. The CPU uses I/O ports to communicate with the keyboard controller, while the KCS interface enables similar communication with the IPMI controller via the system bus.

This is not the name of a driver but rather direct interaction with the I/O.

- **Only support DOS & UEFI**.

The following is an example of usage:

```
C:\>spv_ipmi.exe -I kcs
```

2.3.2.6 h2o

The h2o interface is based on the InsydeH2O(Hardware-to-OS) BIOS. It mainly communicates with BMC through the protocol provided by InsydeH2O BIOS. The BIOS determines which In-Band interface, such as KCS or SSIF(SMBus System Interface), to use for communication with the BMC.

- **Only support UEFI**.

The following is an example of usage:

```
fs0:\>spv_ipmi.efi -I h2o
```

2.3.2.7 open

The open interface utilizes the OpenIPMI kernel device driver. Because this interface requires permission to load the kernel driver, the following example uses 'sudo' to execute with root permissions. Please make sure the **OpenIPMI driver** (in the next section) has been loaded, before running IPMI when using 'open' interface.

The following is an example of usage:

```
./sudo spv_ipmi -I open
```

2.3.2.7.1 How to Load OpenIPMI Driver

Linux: Using Ubuntu as an example:

```
# This provides a userland IOCTL interface for the IPMI driver, each open file for
this device ties into the message handler as an IPMI user.

./sudo modprobe ipmi_devintf

# A driver can support KCS interfaces.

./sudo modprobe ipmi_si
```

FreeBSD:

```
# Load an IPMI driver into the kernel.

./sudo kldload ipmi

# Display status of the dynamic kernel linker.
./sudo kldstat
```

2.3.3 Default

Before each IPMI command is executed, an interface needs to be created to communicate with BMC.

If no interface is specified, it will use KCS by default.

- There are different interfaces for In-Band(local solution) KCS transmission in different OS.

The tool will choose a default KCS interface according to different OS.

- Linux/FreeBSD - Open IPMI interface.
- Windows - MS interface (Microsoft IPMI driver).
- DOS/UEFI - KCS interface.

2.4 Command Structure

For Insyde command instructions, the option must be preceded by the command.

- Usage: spv_ipmi [options...] <command>

The following are the commands extended by Insyde: isc

```
./spv_ipmi -h

...

isc            Insyde Utility Tool
```

The following are the instructions for Insyde isc expansion:

```
Insyde Utility Tool 1.8.18.24301

Insyde Commands:
        bmc            BMC firmware update
        bios           BIOS firmware update
        pld            PLD firmware update
        hpm            Update HPM components using PICMG HPM.1 file
        modmac         Modify MAC address
        chkalert       PEF alert auto test
        sdr            Sensor Data Record
        fru            Field Replaceable Unit
        spd            Serial Presence Detect
        mntiso         Mount Web CD-ROM image
```

```
        version       Tool version
```

Press 'enter' or input 'help' to check how to use any command. Examples are shown below:

```
./spv_ipmi isc bmc help

BMC Commands:
        version       Get BMC version information
        status        Get BMC status
        update        Update BMC firmware
        reset         Reset BMC to idle
        sdkver        Get BMC SDK version
        sdkname       Get BMC SDK name
        cfg           User configuration file
        log           Download debug logs zip file
```

Before each command is executed, a connection to the BMC must be established. The next section will introduce the methods to connect to the BMC.

2.5 Systems Management

The management of server systems and network equipment can be divided into two methods: In-Band(IB) and Out-of-Band(OOB).

2.5.1 In-Band

The In-Band management through KCS is based on connectivity and software that must be installed on the remote system being managed and only works after the operating system is booted.

Prerequisite: The manager must be in the system of the managed device. 'host' must be booted to OS first.

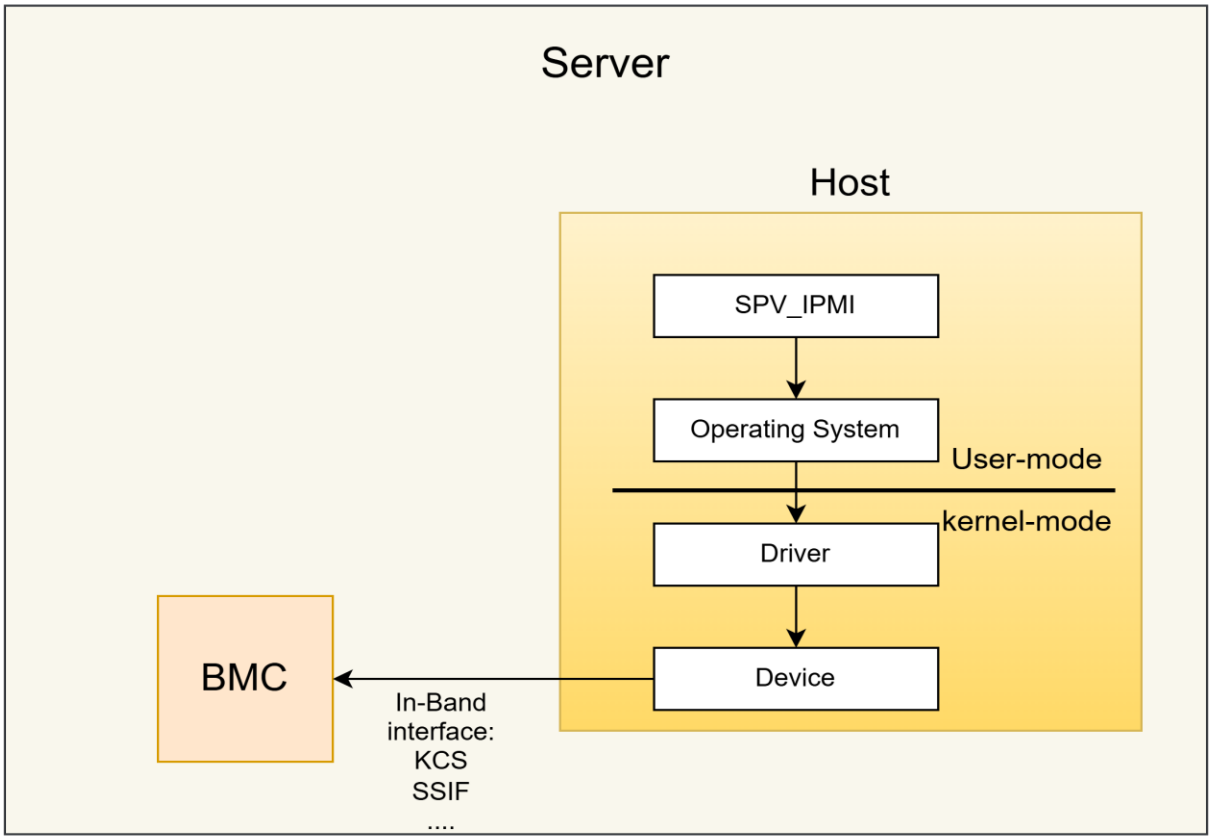

Figure 3-Relationship between BMC and Host

The picture above is the relationship between **BMC** and **Host**.

- **User-mode** is the environment in which applications run, while **kernel-mode** is the environment in which the **OS**(Operating System) interacts directly with the hardware.
- **BMC** is an embedded controller of the server, mainly responsible for hardware management and remote control.
- **Host** is the core part of the **server** and is responsible for running the **OS** and processing user applications.
- Run **SPV_IPMI** on the **OS** on the Host. **SPV_IPMI** will use the System Call Driver of the **OS** to implement **Device** access and communicate with the **BMC.**
- The devices are interface-mapped files, which may be **KCS** or **SSIF**. These device files communicate with the **BMC** through the kernel's IPMI driver.

If no interface is specified, the tool will default to In-Band (refer to the interface section in the help for details, where [default] is indicated). For details, please see the chapter 2.3.3 Default.

When using the UEFI(Unified Extensible Firmware Interface) version of the tool running on InsydeH2O BIOS, consider switching to the h2o interface, which is supported. This interface will utilize the default In-Band management provided by BIOS to communicate with BMC.

Example: In an environment without a provided KCS interface, switching to the h2o interface is well. In this case, the default communication between BIOS and BMC is through SSIF (The SMBus system interface). Therefore, the underlying communication of ipmitool is done through SSIF and BMC.

- Currently, it has been confirmed that the platforms provided by NVIDIA, Starship and Falcon, do not have KCS functionality. Please specify the use of the h2o interface.

The following is an example of usage:

```
./spv_ipmi -I h2o isc
```

2.5.2 Out-of-Band

Out-of-Band management involves the use of remote management interfaces for managing network devices.

Prerequisite: The network channel must be functional, and the network device that running IPMI can connect to BMC.

It is recommended to use lanplus (IPMI v2.0 RMCP+ LAN Interface) to operate because the remote management and control protocol+ (RMCP+) is an encryption network protocol.

The following is an example of the command format:

```
spv_ipmi -I <intf> -H <hostname> -U <username> -P <password> -C <ID> isc <command>
```

The following table is supported parameters:

Table 7-Essential parameters for Out-of-band managment

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc

Insyde Utility Tool 1.8.18.24301

Insyde Commands:
        bmc            BMC firmware update
        bios           BIOS firmware update
        pld            PLD firmware update
```

```
        hpm           Update HPM components using PICMG HPM.1 file
        modmac        Modify MAC address
        chkalert      PEF alert auto test
        sdr           Sensor Data Record
        fru           Field Replaceable Unit
        spd           Serial Presence Detect
        mntiso        Mount Web CD-ROM image
        version       Tool version
```

2.6 ipmitool Commands

ipmitool provides a variety of commands to communicate with BMC, here are examples of firmware update commands.

2.6.1 HPM FW Update

The HPM(Hardware Platform Management) format is associated with firmware updates, specifically used by certain embedded systems and devices, such as those in the context of hardware management and firmware deployment.

For HPM update commands, there is no distinction for firmware because the HPM file will describe which type it currently belongs to. For more information please see https://www.picmg.org/openstandards/hardware-platform-management/.

2.6.2 BMC FW Update

This example shows how to use the ipmitool command 'hpm' to upload the BMC image. With a large buffer of 30000 bytes. Through Out-of-Band(lanplus).

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 -N 10 -z 30000 hpm
upgrade image-update-ast2600.hpm force activate
```

The following table is supported parameters:

Table 8-Supported parameters for BMC firmware update

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
-N	Specifies a timeout of 10 seconds for the IPMI command, only available with the lanplus interface. The default value is 2 seconds	V
-z	Sets the maximum packet size in bytes for data transmission, here is 30000, and is only usable with lanplus	V
upgrade	Copies components from a valid HPM.1 image to the target	
force	Use "force" to bypass the version check results	
activate	Activate the newly uploaded firmware	

The following figure shows the command running result:
Note: The user must wait for the BMC to restart before completing the update (it will take minutes).

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI/build/linux/x64_
release$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 -N 10 -
z 30000 hpm upgrade image-update-ast2600.hpm force activate
Setting large buffer to 30000

PICMG HPM.1 Upgrade Agent 1.0.9:

Validating firmware image integrity...OK
Performing preparation stage...
    Invalid image file for product 256

    Image Information
        Device Id : 0x22
        Prod   Id : 0x0003
        Manuf  Id : 0x00b009
    Board Information
        Device Id : 0x22
        Prod   Id : 0x0100
        Manuf  Id : 0x00b009
 Continue ignoring DeviceID/ProductID/ManufacturingID (Y/N): y

Services may be affected during upgrade. Do you wish to continue? (y/n): y
OK

Performing upgrade stage:

--------------------------------------------------------------------------------
|ID  | Name        |                     Versions                        | %  |
|    |             |      Active     |      Backup     |      File       |    |
|----|-------------|-----------------|-----------------|-----------------|----|
|*  0|BMC          |  35.24 44123700 | ---.-- -------- |  35.24 46124000 |100%|
|    |Upload Time: 01:48             | Image Size: 59217920 bytes                |
--------------------------------------------------------------------------------
(*) Component requires Payload Cold Reset
Performing activation stage:
    Waiting firmware activation...OK

Firmware upgrade procedure successful
```

Figure 4-Screenshot for running BMC firmware update

2.6.3 BIOS FW Update

This example shows how to use the ipmitool command 'hpm' to upload the BIOS image file BHS_0061_SP_WebUpdate.hpm. With force activated and large buffer 30000 configured.

After execution, BIOS will not update immediately. SPV_IPMI will upload the image file to BMC. Then BMC will create a virtual device to mount this file. It is necessary to manually restart the host. When BIOS is restarting, it will find the virtual device and update.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 -z 30000 hpm upgrade
BHS_0061_SP_WebUpdate.hpm force activate
```

The following table is supported parameters:

Table 9-Supported parameters for BIOS firmware update

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
-z	Sets the maximum packet size in bytes for data transmission, here is 30000, and is only usable with lanplus	V
upgrade	Copies components from a valid HPM.1 image to the target	
force	Use "force" to bypass the version check results	
activate	Activate the newly uploaded firmware	

The following figure shows the command running result:

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI/build/linux/x64_
release$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 -N 10 -
z 30000 hpm upgrade BHS_0061_SP_WebUpdate.hpm force activate
Setting large buffer to 30000

PICMG HPM.1 Upgrade Agent 1.0.9:

Validating firmware image integrity...OK
Performing preparation stage...
    Invalid image file for product 256

    Image Information
        Device Id : 0x22
        Prod   Id : 0x0003
        Manuf  Id : 0x00b009
    Board Information
        Device Id : 0x22
        Prod   Id : 0x0100
        Manuf  Id : 0x00b009
 Continue ignoring DeviceID/ProductID/ManufacturingID (Y/N): y

Services may be affected during upgrade. Do you wish to continue? (y/n): y
OK
Performing upgrade stage:

-------------------------------------------------------------------------------
|ID  | Name        |                     Versions                        | %  |
|    |             |      Active     |      Backup     |      File       |    |
|----|-------------|-----------------|-----------------|-----------------|----|
|*  2|BIOS         |   0.00 00000000 | ---.-- -------- |   3.52 00170013 |100%|
|    |Upload Time: 00:35             | Image Size: 19696511 bytes              |
-------------------------------------------------------------------------------
(*) Component requires Payload Cold Reset
Performing activation stage:
    Waiting firmware activation...OK

Firmware upgrade procedure successful
```

Figure 5-Screenshot for running BIOS firmware update

The following figure shows the updating screen by KVM(Kernel-based Virtual Machine):

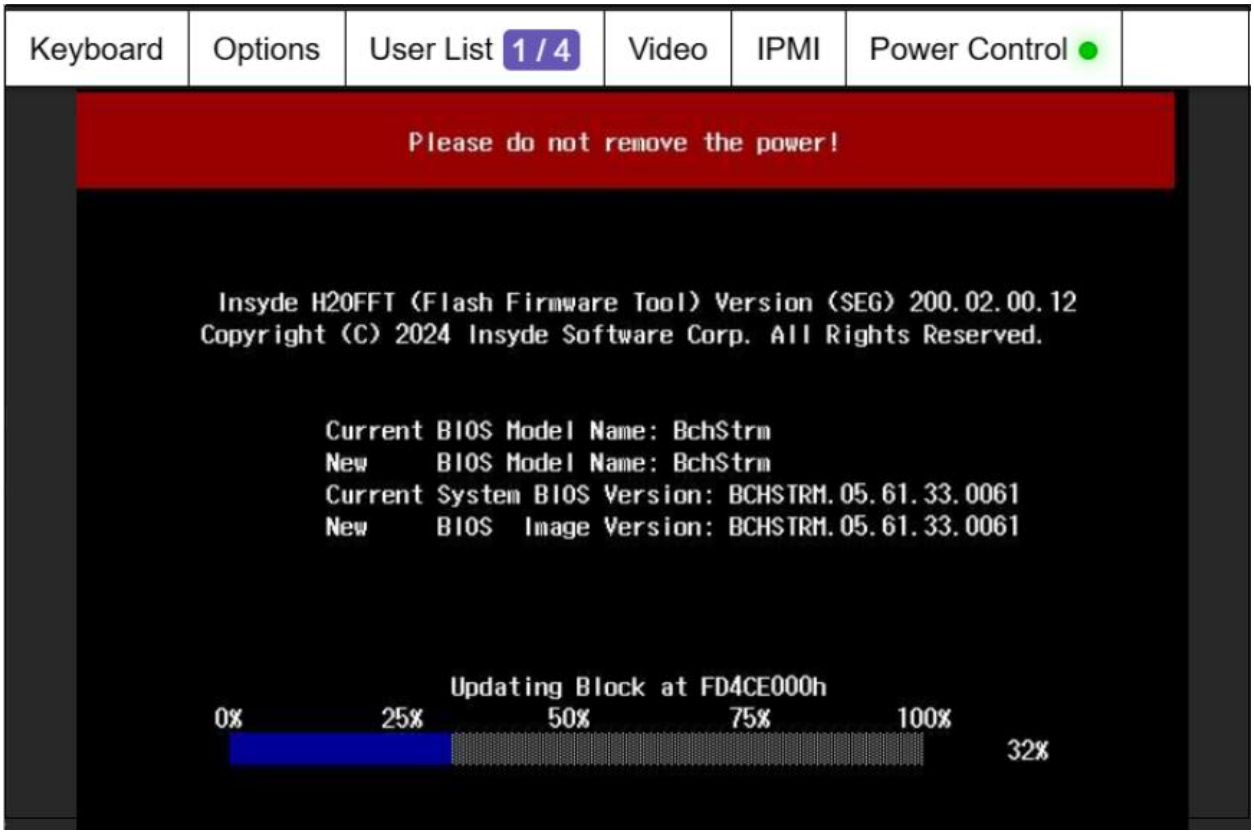

Figure 6-Screenshot from KVM for running BIOS firmware update

2.7 Insyde Commands

The following is an introduction to using Insyde OEM Command and Supervyse OPF BMC for communication.

The examples provided are all about communicating with BMC through remote access on Ubuntu. Please refer to the previous instructions to switch to In-Band.

The error message is shown as follows when the wrong interface is input:

```
./spv_ipmi -I lanplus2 -H 192.168.9.172 -U root -P OpenBmc -C 17 isc

Error loading interface lanplus2
```

Once the wrong cipher suite ID, IP address, user name, or password is inputted, the error message is listed as follows:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 16 isc

Error in open session response message : invalid confidentiality algorithm
Error: Unable to establish IPMI v2 / RMCP+ session

./spv_ipmi -I lanplus -H 192.168.9.172 -U root2 -P OpenBmc -C 17 isc

Error: Unable to establish IPMI v2 / RMCP+ session
```

The following table is the description of the command return value:

Table 10-Description of return value for isc sub-commands

Return Value	Description
0	Success
-1	Error
-2	Unsupported
57	Invalid parameter

2.7.1 Version

The version number of SPV_IPMI can be retrieved by the command as follows:

```
./spv_ipmi isc version

Insyde Utility Tool 1.8.18.24301
```

The following table is version number format:

Table 11-Version number format

Description	Version	Explain
ipmitool Version	1.8.18	Base ipmitool v1.8.18

Description	Version	Explain
Year	24	2024 year
Quarter	3	Q3
Build Number	01	First release of Q1

2.7.2 Insyde HPM FW Update

Due to the limited transmission efficiency of the In-Band 'hpm' command provided by ipmitool, the speed is very slow. Insyde uses BMC-VM(Virtual Media) to simulate a virtual device and places the image file in the virtual device to overcome the problem of upload performance.

Note： HPM is not only for BMC but also BIOS.

The following is an example of usage:

```
sudo ./spv_ipmi isc hpm upgrade image-update-ast2600.hpm usb force
```

The following table is supported parameters:

Table 12-Supported parameters for Insyde HPM FW Update

Parameter	Description	Optional
upgrade	Copies components from a valid HPM.1 image to the target	
usb	Simulate a virtual device as a USB device	
force	Use "force" to bypass the version check results	

The following figure shows the command running result:

```
Get firmware update key.o ./spv_ipmi isc bmc update image-update-ast2600.hpm usb
Enter firmware update mode.
USB attach.
Set file name.
-->usb://image-update-ast2600.hpm
Wait for USB connection.

Copy File to USB DEV.
Mount a file systems.
File Name       : /media/IPMI_BMC_USB/image-update-ast2600.hpm
Upload file to USB successfully!
Unmount file systems.
USB detach.

Status: Programming Image (0x04)
100 % [########################################]
Status: Ready - Successfully Update (0x05)

Ready for reboot.
```

Figure 7-Screenshot to running HPM fw update

2.7.3 BMC

This section describes BMC-related functions.

The following are related commands:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc bmc

BMC Commands:
        version       Get BMC version information
        status        Get BMC status
        update        Update BMC firmware
        reset         Reset BMC to idle
        sdkver        Get BMC SDK version
        sdkname       Get BMC SDK name
        cfg           User configuration file
        log           Download debug logs zip file
```

2.7.3.1 Firmware Update Flow

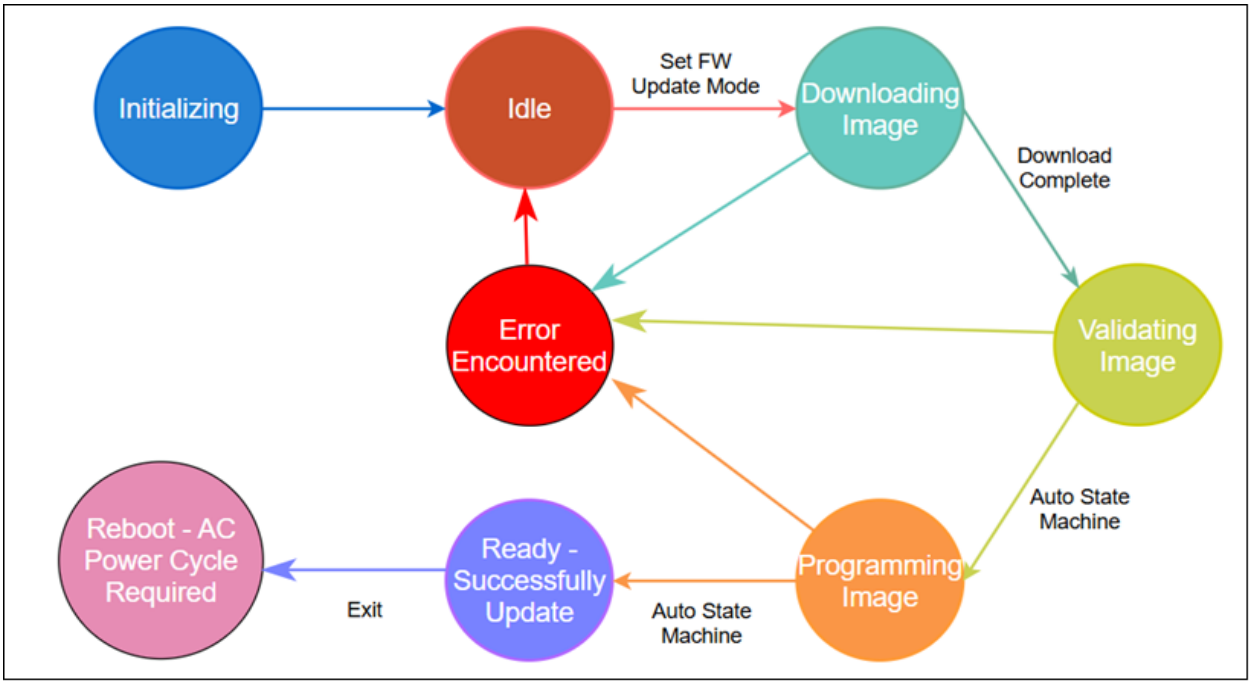

Figure 8-BMC Firmware Update Stages

The BMC firmware update flow contains 5 stages

Idle: Enter firmware update mode before performing the firmware update.

Downloading Image: Upload the BMC firmware image.

Validating Image: The transfer is complete. Validating the BMC firmware image just uploaded.

Programming Image: Validation is complete, programming the BMC firmware image.

Ready: Programming is done, reset BMC and boot from the new image.

2.7.3.2 Get BMC Version

Get the version number of the BMC.

SPV_IPMI provides commands to query the BMC version. It includes the versions of BMC, Boot-Block-Updateable (BBU), and Boot-Block-ROM (BBR). The default is BMC Active Image.

OPF only supports **BMC Active Image** currently.

Note: Whether the function is available depends on whether BMC supports it.

The following is an example of help:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc bmc version help

usage: version <FW-TYPE>
    FW type:
      0x01 - BMC Active Image (Default)
      0x02 - BBU Active Image
```

```
      0x03 - BMC Backup Image
      0x04 - BBU Backup Image
```

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc version
```

The following table is supported parameters:

Table 13-Supported parameters for BMC version

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc version

Type          : BMC Active Image

Version       : 35.18.2e0004d8
```

2.7.3.3 Get BMC Status

Only the idle state allows updates, so please check the current BMC status before updating.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc status
```

The following table is supported parameters:

Table 14-Supported parameters for BMC status

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	

Parameter	Description	Optional
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc bmc status

Status                : Idle
Percentage            : 0
Hash Check            : None requested
```

The following table shows possible return values:

Table 15-Possible return value for BMC status

Output	Available Value	Description
Status	Initializing	BMC is initializing
	Idle	BMC is idle
	Downloading Image	BMC is busy downloading the image
	Validating Image	BMC is busy validating the image
	Programming Image	BMC is busy programming the image
	Ready - Successfully Update	BMC is updating the image successful
	Error Encountered	When there is an error, the FW update error code and error message obtained from the BMC will be displayed.
	Update Forbidden by Policy	BMC update disabled
	AC Power Cycle Required	Need to do AC power cycle
Percentage	Percentage number	Percentage of completion of BIOS image validating or programming
Hash Check	None requested	No hash check requested
	Check requested	Hash check is requested
	Passed SHA2	SHA2 check pass
	Passed CRC32	CRC32 check pass
	Failed SHA2	SHA2 check failed
	Failed CRC32	CRC32 check failed

2.7.3.4 Reset BMC Update Mode

Set the BMC update mode to Idle. BMC needs to be reset when previous updating fails, which can reduce the waiting time for the user.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc reset
```

The following table is supported parameters:

Table 16-Supported parameters for BMC reset

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc reset

Reset BMC update mode to IDLE.
```

The following table shows possible return values:

Table 17-Possible return values for BMC reset

Output	Description
Reset BMC update mode to IDLE	Successfully set BMC update mode to idle
Reset BMC update mode failed	Adjust the BMC update mode failed

2.7.3.5 Get BMC SDK Version

Get the BMC SDK(Software Development Kit) version number.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc sdkver
```

The following table is supported parameters:

Table 18-Supported parameters for BMC SDK Version

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	

Parameter	Description	Optional
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc sdkver

SDK version: 35.24.46.1240
```

The following table is the version format:

Table 19-Version format descriptions

Version	explain
35	3 is OPF Product Type (0: reserved for Module, 1: SPF, 2: PM, 3: OPF)
	5 is Major Version
24	Year. 24 means 2024
46	Release work week number
1240	1 is Intel Platform ID (0:All, 1:Intel, 2: AMD, 3:NVIDIA, 4: Ampere 9: EVB)
	2 is Pre-Release state. (0: Formal Release 2: Pre-release for testing)
	40 is Release Counter, 00-99, increase 1 every release, reset every year

2.7.3.6 Get BMC SDK Name

Get the BMC SDK name.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bmc sdkname
```

The following table is supported parameters:

Table 20-Supported parameters for SDK Name

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	

Parameter	Description	Optional
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc bmc sdkname

Supervyse OPF RV24.12.4
```

The following table is the version format:

Table 21-Version format for SDK Name

Version	explain
24	Year. 24 means 2024
12	Month
4	Sequential numbering

2.7.3.7 BMC FW Update

Use Insyde IPMI OEM Command to update the BMC firmware. The following is notable when executing the command:

- Use the HPM file to update the BMC Image through Insyde Command.
- In OPF BMC, only HPM file format is supported.
- Before updating, the user must check whether the BMC version is correct, and the BMC status is idle.
- In this command, what SPV_IPMI does is only upload the file, and BMC will restart itself to complete the update process.

Note: The update process will take several minutes. Taking Intel BHS as an example, BMC will automatically restart about five minutes after SPV_IPMI execution.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.9.172 -C 17 isc bmc update
image-update-ast2600.hpm
```

The following table is supported parameters:

Table 22-Supported parameters for BMC FW update

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	

Parameter	Description	Optional
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following figure is the command running result:

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI/build/linux/x64_
release$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc bmc
 update image-update-ast2600.hpm
Firmware upload start.
Image size      : 56.43 (mb)
100 % [########################################]
Firmware upload complete.

Get firmware update key.
Enter firmware update mode.
Set file name.

Status: Programming Image (0x04)
100 % [########################################]
Status: Ready - Successfully Update (0x05)

Ready for reboot.
```

Figure 9-Screenshot for Insyde BMC FW Update

2.7.3.8 BMC FW Update via USB

Due to the limited transmission efficiency of the In-Band command provided by ipmitool, the speed is very slow. Insyde uses BMC-VM(Virtual Media) to simulate a virtual USB device and places the image file in the virtual device to overcome the problem of upload performance.

Note: It also uses In-Band commands (such as KCS or SSIF, etc.) to send IPMI command requests to the BMC for creating virtual devices, rather than using the USB interface.

The following is an example of usage:

```
sudo ./spv_ipmi isc bmc update image-update-ast2600.hpm usb
```

The following table is supported parameters:

Table 23-Supported parameters for Insyde BMC fwupdate through USB

Parameter	Description	Optional
usb	Simulate a virtual device as a USB device	

The following figure is the command running result:

```
Get firmware update key.o ./spv_ipmi isc bmc update image-update-ast2600.hpm usb
Enter firmware update mode.
USB attach.
Set file name.
-->usb://image-update-ast2600.hpm
Wait for USB connection.

Copy File to USB DEV.
Mount a file systems.
File Name        : /media/IPMI_BMC_USB/image-update-ast2600.hpm
Upload file to USB successfully!
Unmount file systems.
USB detach.

Status: Programming Image (0x04)
100 % [##########################################]
Status: Ready - Successfully Update (0x05)

Ready for reboot.
```

Figure 10-Screenshot for Insyde BMC fwupdate through USB

The following is the KVM screen. It uses In-Band mode to run commands on the host, and the user can watch the result through the KVM screen.

```
insyde@insyde-bhs:~$ sudo ./spv_ipmi  isc bmc update image-update-ast2600.hpm usb
[sudo] password for insyde:
Get firmware update key.
Enter firmware update mode.
USB attach.
Set file name.
-->usb://image-update-ast2600.hpm
Wait for USB connection.

Copy File to USB DEV.
Mount a file systems.
File Name        : /media/IPMI_BMC_USB/image-update-ast2600.hpm
Upload file to USB successfully!
Unmount file systems.
USB detach.

Status: Programming Image (0x04)
100 % [##########################################]
Status: Ready - Successfully Update (0x05)

Ready for reboot.
insyde@insyde-bhs:~$
```

Figure 11-Screenshot from KVM for running Insyde FW update

2.7.3.9 Backup / Restore User Configuration File

BMC settings can be backed up to a file, and the settings can be restored to the BMC or other BMCs through the file.

The user configuration includes network configuration, IPMI channel configuration, VM/IKVM configuration, Redfish configuration, BMC-web configuration, BIOS configuration, SNMP configuration, and third-party authentication configuration...etc.

2.7.3.9.1 Backup User Configuration File

Example: Backup user configuration File to 'backup.bin'.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.9.172 -n 900 isc bmc cfg
backup backup.bin
```

The following table is supported parameters:

Table 24-Supported parameters for Backup user configuration files

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
-n	Set packet size (bytes) for firmware update. The response packet header includes OpenSSL, which will occupy the actual content, so it needs to be dynamically adjusted	V

The following figure is the command running result:

The following table shows possible return values:

```
insyde@insyde-virtualbox-2004:~/Downloads$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.
.172 -C 17 -n 900 isc bmc cfg backup backup.bin
Backup user configuration file.

Download Size: 197 KB
100 % [########################################]

Success to download configuration file.
```

Figure 12-Screenshot for running backup user configuration file

Table 25-Possible return values for backup user configuration file

Response Message	
Timed out - over 20 second	The BMC configuration file has not been backed up after 20 seconds.
Failed to get configuration file size	There is no way to know the size of the BMC configuration file. SPV_IPMI needs to know the size before downloading it.

Response Message	
Failed to create file name	Unable to create the local file.
Success to download configuration file	Successfully downloaded the configuration file from BMC.

2.7.3.9.2 Restore User Configuration File

To restore the user configuration file, please follow the steps below:

- A backup file containing BMC settings is necessary.
- The restore procedure must wait for the BMC to restart to take effect.

The following is an example of usage to restore user configuration from file 'backup.bin':

```
./spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.9.172 isc bmc cfg restore
backup.bin
```

The following table is supported parameters:

Table 26-Supported parameters for restore user configuration file

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
force	Use "force" to bypass the version check results	V

The following figure is the command running result:

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$
/spv_ipmi -I lanplus -U root -P OpenBmc -C 17 -H 192.168.9.172  isc bmc cfg res
ore  backup.bin

Note: BMC will automatically restart after the restoration is successful!! (Y/N
:
y
Start upload user configuration file.
Transfer user configuration file.
100 % [########################################]
Finish upload user configuration file.
Verification : Pass

Restore user configuration file.
Please wait for the process.
Restart BMC.

Success to restore user configuration file.
```

Figure 13-Screenshot for running restore user configuration file

Note: BMC will verify whether the file is valid, restore the settings, and restart BMC.

The following message will be displayed when an error occurs:

Table 27-Possible output for restore user configuration file

Possible Value	Description
Unable to find file: XXX	The BMC configuration file could not be found

The following message will be displayed during command execution:

Table 28-Descriptions to messages during restore user configuratoin file

Possible Value	Description
Start upload user configuration file	Send the upload command to the BMC
Transfer user configuration file	Start transferring configuration files to the BMC
Finish upload user configuration file	Send the upload completed command to the BMC
Verification: Pass	The BMC verifies that the configuration file just uploaded is valid
Verification: Failed	The BMC verifies that the configuration file just uploaded is invalid
Restore user configuration file	Send a restore configuration file command to the BMC
Restart BMC	Send a command to restart the BMC to the BMC
Success to restore user configuration file	SPV_IPMI has completed the restoration of the BMC configuration file

The following figure is the force command running result. Because the 'force' option is input, SPV_IPMI will not ask the user to restart BMC.

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI_BIN/linux/x64$ .
/spv_ipmi -I lanplus -U root -P 0penBmc -C 17 -H 192.168.9.172  isc bmc cfg rest
ore  backup.bin force
Start upload user configuration file.
Transfer user configuration file.
100 % [########################################]
Finish upload user configuration file.
Verification : Pass

Restore user configuration file.
Please wait for the process.
Restart BMC.

Success to restore user configuration file
```

Figure 14-Screenshot to Force restore user configuration

2.7.4 BIOS

This section describes about BIOS FW(firmware) update via SPV_IPMI.

The following are related commands:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios

BIOS Commands:
        status        Get BIOS status
        update        Update BIOS firmware
        reset         Reset BIOS firmware to IDLE
        ccmos         Clear CMOS
        download      Download BIOS firmware
```

2.7.4.1 Get BIOS status

This function is to check how the BMC updates the BIOS.

BMC offers multiple methods for updating the BIOS, and each method corresponds to its own specific status.

Only the idle state allows updates, check the current BIOS update status before updating.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios status
```

The following table is supported parameters:

Table 29-Supported parameters for Insyde Command-Get BIOS Status

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

Intel updates the BIOS via virtual media. This example is executed on Intel BHS.

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios status
```

```
BIOS Update Method    : Virtual Media
BIOS update mode      : No
Image uploaded        : No
Virtual USB ready     : No
```

Note: 'No' means idle.

The following table is the possible values:

Table 30-Possible value for BIOS Status

Output	Available Value	Description
BIOS Update Method	Virtual Media	BMC updates BIOS through Virtual Media
	SPI Controller	BMC updates BIOS through SPI(Serial Peripheral Interface)
	PLDM	BMC updates BIOS through PLDM(Platform Level Data Model)

When the BMC updates BIOS via Virtual Media:

Table 31-Possible value for BIOS Status when updated via Virtual Media

Output	Available Value	Description
BIOS update mode	Yes	Currently entering BIOS update mode
	No	Not currently in BIOS update mode
Image uploaded	Yes	The BIOS firmware image has been uploaded to the BMC
	No	The BIOS firmware image has not been uploaded to the BMC yet
Virtual USB ready	Yes	USB virtual disk is ready for BIOS FW UPDATE
	No	USB virtual disk is not ready for BIOS FW UPDATE
Validating	Percentage value	BIOS is validating the new firmware image
Programming	Percentage value	BIOS is programming the new firmware image
Status	Update success	Upload successful
	Update init failed	Initialization failed
	Validate image failed	BIOS Image verification failed
	Program image failed	BIOS Image programming failed

When the BMC updates the BIOS, it reads the current mode of the SPI(Serial Peripheral Interface) controller before the BIOS update through SPI or PLDM(Platform Level Data Model).

Table 32-Possible output on updating BIOS through SPI/PLDM

Output	Available Value	Description
SPI Mode	Disable SPI interface	Disable SPI interface mode
	Enable SPI Master	Enable SPI master mode
	Reserved, enable SPI Master and SPI Slave to AHB Bridge (debug mode)	Enable SPI master and SPI slave mode
	Enable SPI Pass-through	Enable SPI pass-through mode
BIOS Update Status	Idle	Idle
	Programming	Updating, the percentage will be displayed during the update

Output	Available Value	Description
	Program Done	Update completed

If the BIOS supports PFR(Platform Firmware Resiliency), there are the following possible output.
To get a more detailed explanation, please refer to '606686_Intel_PFR_ Arch_HW_Guide_Rev_2_5.pdf' from Intel.

Table 33-Possible output on updating BIOS with PFR supported

Output	Possible Value
CPLD RoT Release Version	Version
CPLD RoT SVN	Number
Platform state	CPLD NIOS II processor waiting to start
	CPLD NIOS II processor started
	Enter T-1
	T-1 Reserved
	BMC flash authentication
	PCH flash authentication
	Lockdown due to authentication failure
	Enter T0
	T0 BMC booted
	T0 ME booted
	T0 ACM booted
	T0 boot completed
	PCH FW update
	BMC FW update
	CPLD update
	CPLD update in recovery mode
	PCH/CPU firmware volume update
	PCH/CPU firmware volume update completed
	T-1 FW recovery
	T-1 forced active FW recovery
	WDT timeout recovery
	CPLD recovery in recovery mode
	Lockdown due to PIT L1
	PIT L2 FW sealed
	Lockdown due to PIT L2 PCH hash mismatch
	Lockdown due to PIT L2 BMC hash mismatch
Recovery count	Number
Last Recovery reason	Normal
	PCH active failure
	PCH recovery failure
	ME launch failure

Output	Possible Value
	ACM launch failure
	IBB launch failure
	OBB launch failure
	BMC active failure
	BMC recovery failure
	BMC launch failure
	CPLD WDT expired
	Forced active FW recovery
Panic event count	Number
Last Panic reason	Normal
	CPLD WDT expired
	BMC WDT expired
	ME WDT expired
	ACM WDT expired
	IBB WDT expired
	OBB WDT expired
	BMC active authentication failure
	BMC recovery authentication failure
	PCH active authentication failure
	PCH recovery authentication failure
	IBB authentication failure
	OBB authentication failure
	PCH update intent
	BMC update intent
	BMC reset detected
Major error code	Normal
	BMC authentication failure
	PCH authentication failure
	BMC boot failure
	ME boot failure
	ACM boot failure
	BIOS boot failure
	PCH update failure
	BMC update failure
	CPLD update failure

2.7.4.2 Reset BIOS Update Mode

Exit BIOS firmware update mode.

If there is any problem during the BIOS update process, use this command to reset the BIOS status to idle.

The following is an example of usage:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios reset
```

The following table is supported parameters:

Table 34-Supported parameters for resetting BIOS update

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios reset

Current is exit BIOS firmware update mode.
```

The following table is the possible values:

Table 35: Possible value for resetting BIOS update

Response Message
Success to exit BIOS firmware update mode.
Failed to exit BIOS firmware update mode.
Current is exit BIOS firmware update mode.

2.7.4.3 Update BIOS Firmware

First shut down the host before updating BIOS.

An HPM image file is necessary.

Note: The following uses Insyde BIOS via virtual media as an example.

The following is an example for Intel BHS:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc bios update
BHS_0061_SP_WebUpdate.hpm
```

The following table is supported parameters:

Table 36-Supported parameters for update BIOS firmware

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following figure is the command running result:

```
insyde@insyde-VirtualBox:/media/sf_Work/Work_Git/Supervyse_IPMI/build/linux/x64_
release$ ./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc bio
s update BHS_0061_SP_WebUpdate.hpm

Current is exit BIOS firmware update mode.

Firmware upload start.
Image size      : 18.78 (mb)
100 % [########################################]
Firmware upload complete.

Enter BIOS update mode
Waiting Virtual USB ready for BIOS to start FW update...
Virtual USB is ready
BIOS will be updated until reboot next time!
```

Figure 15-Screenshot for running BIOS firmware update

After execution, it will not update immediately. It will create a virtual device with a BIOS image inside. Restarting the host and BIOS will find the virtual device and update the BIOS.

The following figure is the firmware updating screen:

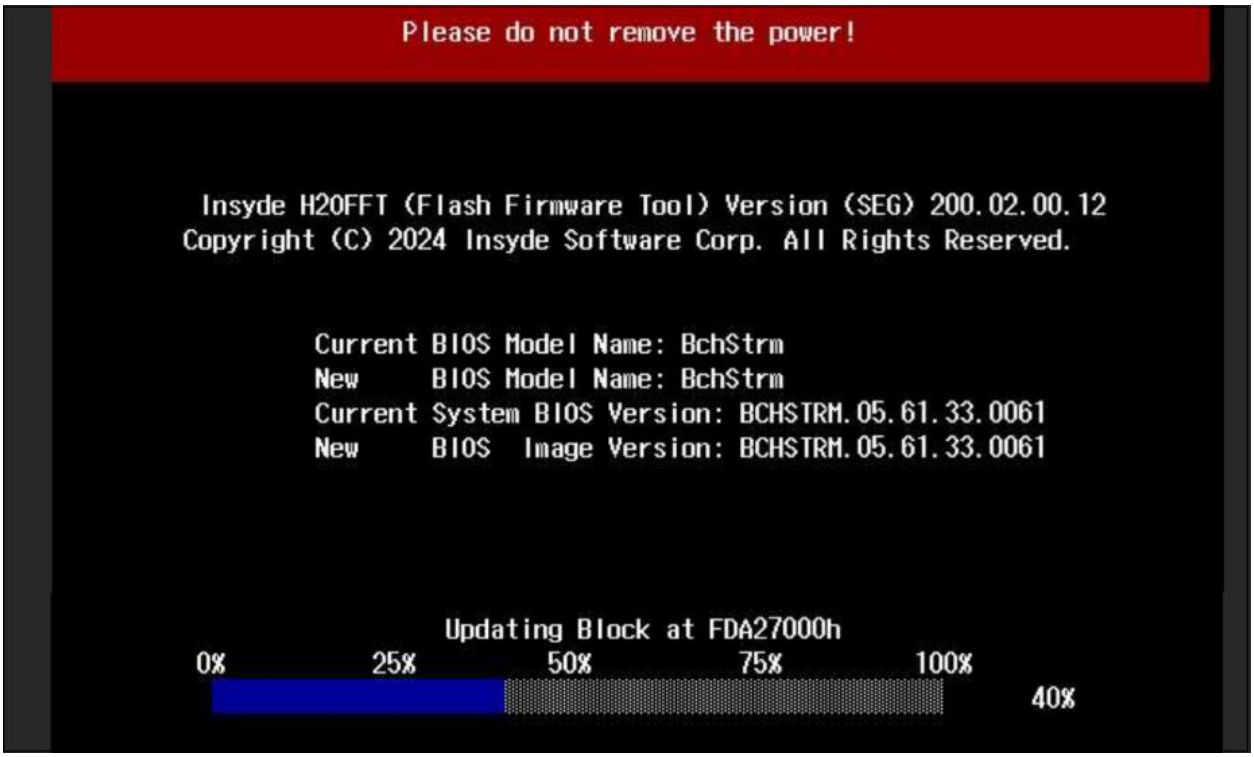

Figure 16-Screenshot from KVM for running BIOS update

2.7.4.4 Clear CMOS

Clear CMOS(Complementary Metal Oxide Semiconductor) is only implemented on specific platforms and is related to hardware configuration.

BIOS is used to identify and configure hardware in the computer. For no boot or display issues, clearing CMOS may help recover the boards because that restores the BIOS default settings.

- Currently only supported on certain platforms (such as AMD).

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc bios ccmos
```

The following table is supported parameters:

Table 37-Supported parameters for clear CMOS

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	

Parameter	Description	Optional
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios ccmos

Not support current platform!!

Failed to clear CMOS!!
```

2.7.4.5 Download BIOS Firmware

When the platform supports updating the BIOS through SPI, it supports downloading the BIOS firmware Image.

Note: Using the command 'status' can get the current BIOS update method.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc bios download
```

The following table is supported parameters:

Table 38-Supported parameters for download BIOS firmware

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc bios download

Download function only supports SPI mode!!
```

The following table is the possible values:

Table 39-Possible values for download BIOS

Output	Available Value	Description
	Download function only supports SPI mode!!	Download function only supports SPI mode
	BIOS does not support download process.	The current BMC does not support the BIOS download process
	Failed to get flash size!	Unable to get BIOS size
BIOS Download Mode	Idle	BIOS download mode is currently idle
	Reading flash is going on	BMC is reading BIOS data
	Program Done	Completed reading BIOS data
Downloading	Percentage value	
Error	Please to wait BMC idle	Please wait for the BMC side BIOS download mode to be idle before starting to download the BIOS file
	Failed to create file name	Unable to write file

2.7.5 Other

2.7.5.1 Modify MAC address

OPF does not yet support it.

It is recommended to operate the following command in the In-Band mode.

The following are related commands:

```
./spv_ipmi isc modmac

usage: modmac chan mac [norest] [pwd]
  chan <number>   - Channel number(1~17)
  mac <address>   - MAC address(format 02:01:02:53:74:1c)
  noreset         - No Reset BMC
  pwd <pass code> - Password (Max is 16 bytes)
```

The following is an example of usage:

```
./spv_ipmi isc modmac 1 mac 01:02:03:04:05:06
```

2.7.5.2 SDR Update

Sensor data record (SDR) is the records that IPMI includes. They include software information, such as how many sensors are present, what type they are, their events, threshold information, and so on.

Currently, **OPF does not support updating SDR via *.sdr files**.

The following are related commands:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc sdr

SDR Commands:
      update        Update SDR information
      status        Get SDR auto configuration status
      parse         Set SDR auto configuration parse control
      read          Read SDR auto configuration file
```

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.9.172 -C 17 isc sdr update
FCT.sdr
```

2.7.5.3 FRU Update

FRU(Field Replacement Unit) is the name for the BMC's non-volatile storage region containing descriptions of the system components that can be serviced as a single entity. This is intended to help the service technical identify the serviceable entity in the managed system. SPV_IPMI supports updating FRU conveniently.

Note: FRU may have 'write' protection. If updating fails, ask the hardware-related personnel.

The following are related commands:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc fru

fru commands:
  update   - update FRU information
```

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc fru update 0
FRU.fru
```

The following table is the possible values:

Table 40-Possible values for FRU update

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
FRU_ID	FRU ID, here is '0'	
FRU_FILE	The file extension is *.fru, here is 'FRU.fru'	

The following figure is the command running result:

To check the result of the FRU update, please enter the following command:

```
insyde@insyde-virtualbox-2004:~/Downloads$ ./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17
sc fru update 0 FRU.fru
Fru Size          : 512 bytes
Size to Write     : 104 bytes
```

Figure 17-Screenshot for FRU update

```
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.9.172 -C 17 fru print 0
```

2.7.5.4 Mount ISO

Insyde provides a way to share ISO with BMC. Use the BMC virtual media function to mount the Web ISO image as an operating system.

Web ISO is an ISO image that can be used to boot host via BMC web.

The following are related commands:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc mntiso

mntiso Commands:
        status        Get VM status and type
        get           Get Web ISO host, path and username
        set           Set Web ISO host, path, username and password
        mount         Mount Web CD-ROM image
        umount        Unmount Web CD-ROM image
```

2.7.5.4.1 How to set an ISO for mounting via NFS

Here is an example of creating a Web ISO path via NFS. NFS(Network File Sharing) is a protocol that defines the way files are stored and retrieved from storage devices across networks. In this example, the operating system is **Ubuntu**. Then install the NFS server by the command as follows:

```
sudo apt install nfs-kernel-server
```

Add one line to file '/etc/exports '.

```
# This is the ISO image file path.

/home/insyde/Documents/ *(ro)
```

Then restart the NFS server.

```
 sudo systemctl restart nfs-kernel-server
```

Check the status of the NFS server by the command as follows:

If the status of the server is active(Active field), SPV_IPMI can mount this ISO file now.

```
insyde@insyde-virtualbox-2004:~$ sudo systemctl status nfs-kernel-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/lib/systemd/system/nfs-server.service; enabled; vendor preset: enabled)
     Active: active (exited) since Tue 2024-12-24 14:40:43 CST; 24s ago
    Process: 7186 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SUCCESS)
    Process: 7187 ExecStart=/usr/sbin/rpc.nfsd $RPCNFSDARGS (code=exited, status=0/SUCCESS)
   Main PID: 7187 (code=exited, status=0/SUCCESS)

┌─ 24 14:40:42 insyde-virtualbox-2004 systemd[1]: Starting NFS server and services...
├─ 24 14:40:42 insyde-virtualbox-2004 exportfs[7186]: exportfs: /etc/exports [3]: Neither 'sub>
├─ 24 14:40:42 insyde-virtualbox-2004 exportfs[7186]:   Assuming default behaviour ('no_subtre>
├─ 24 14:40:42 insyde-virtualbox-2004 exportfs[7186]:   NOTE: this default has changed since n>
└─ 24 14:40:43 insyde-virtualbox-2004 systemd[1]: Finished NFS server and services.
lines 1-12/12 (END)
```

Figure 18-Check NFS Status by systemctl status

2.7.5.4.2 Get VM Mount information

This section describes how to get current virtual device information.
If this parameter 'device index' is omitted, SPV_IPMI will display the range of device indexes.

The following is an example to get the range of device index:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc mntiso status

usage: status <device_index>
  device_index - device index [1-5]
```

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc mntiso status 1
```

The following table is the Supported parameters:

Table 41-Supported parameters for Getting VM Mount Information

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
Device index	Here is '1'	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc mntiso status 1

Media Type         : Unused
```

```
Connected Via      : Not connected
Inserted Status    : FALSE
Read Only          : TRUE
```

The following table is the possible values:

Table 42-Possible values for running Get VM Mount Information

Output	Possible Value	Description
Media Type	Unused	BMC has not mounted any ISO image
	Floppy	The media type is a Floppy
	USB stick	The media type is a USB stick
	CDROM	The media type is a CD-ROM
Connected Via	Not connected	BMC has not mounted any ISO image
	URI	Connected via URI(Uniform Resource Identifier)
	Applet	Connected via Java Applet
	OEM	Connected via IPMI OEM commands
Inserted Status	FALSE	Inserted status is false
	TRUE	Inserted status is true
Read Only	FALSE	`Write protected is false`
	TRUE	`Write protected is true`

2.7.5.4.3 Set Web ISO information

Setting this information before mounting ISO from a remote server is necessary.

Since the connection process of the Out-of-Band interface already includes the password and account of BMC, there is no need to specify them separately.However, in In-Band mode, a specify password and account of BMC is necessary.

In this example, the ISO file path is /home/insyde/Documents/ubuntu2004.iso, and NFS server IP is 192.168.9.20.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc mntiso set 1 nfs
192.168.9.20 /home/insyde/Documents/ubuntu2004.iso
```

The following table is the supported parameters:

Table 43-Supported parameters for Set Web ISO Information

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	

Parameter	Description	Optional
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
Device index	Here is '1'	
Share type	'samba' or 'nfs'	
Host name	Web ISO hostname or IP address, here is '192.168.9.20'	
Image path	Web ISO path(Support .iso, .img or .ima file, and folder '/')	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc mntiso set 1 nfs
192.168.9.20 /home/insyde/Documents/ubuntu2004.iso

Success to Set share host: '192.168.9.20'
Success to Set path to Image: '/home/insyde/Documents/ubuntu2004.iso'
```

The following table is possible values:

Table 44-Possible values for set Web ISO Information

Possible Value
`Success to Set share host: '192.168.9.20'`
`Failed to Set share host: '192.168.9.20'`

2.7.5.4.4 Get Web ISO information

This command can get relevant information if Web ISO has been set.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc mntiso get 1
```

The following table is the supported parameters:

Table 45-Supported parameters for Get Web ISO Information

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	

Parameter	Description	Optional
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
Device index	Here is '1'	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc mntiso get 1

Share Host      : 192.168.9.20
Path To Image   : /home/insyde/Documents/ubuntu2004.iso
```

The following table is possible values:

Table 46-Possible values for Set Web ISO Information

Possible Value
`Fail to get VM web ISO configuration`
`Failed to allocate memory block`
`Unable to find title: xxx`

2.7.5.4.5 Mount Web CD-ROM image

This command can mount the Web ISO.

Set Web ISO information must be completed before running this command.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P 0penBmc -H 192.168.9.172 -C 17 isc mntiso mount 1
```

The following table is the supported parameters:

Table 47-Supported parameters for Mount Web CD-ROM image

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	

Parameter	Description	Optional
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
Device index	Here is '1'	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc mntiso mount 1

Success to mount image to device 1
```

The mount status can be watched as follows:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P OpenBmc -C 17 isc mntiso status 1

Media Type         : CDROM
Connected Via      : URI
Inserted Status    : TRUE
Read Only          : TRUE
```

The following table is possible values:

Table 48-Possible values for running Mout Web CD-ROM image

Possible Value
Device index is invalid
Fail to get VM web ISO configuration
Host is invalid
Path is invalid
Fail to mount web ISO
Failed to allocate memory block
User name or password length over 32 bytes
Failed to get VM status
Device has been inserted
Success to mount image to device 1
Failed to mount image to device 1

2.7.5.4.6 Unmount Web CD-ROM image

This command 'umount' unmounts the mounted Web ISO image.

The following is an example of usage:

```
./spv_ipmi -I lanplus -U root -P OpenBmc -H 192.168.9.172 -C 17 isc mntiso umount 1
```

The following table is the supported parameters:

Table 49-Supported parameters for unmount Web CD-ROM image

Parameter	Description	Optional
-I	Specifies the lanplus communication interface, which supports IPMI v2.0	
-U	Sets the username for IPMI login, here is 'root'	
-P	Sets the password for IPMI login, here is '0penBmc'	
-H	Defines the target IP address of the IPMI-enabled device, here is '192.168.9.172'	
-C	This option sets the cipher suite ID for encryption, with 17 being the selected suite. This option is specific to lanplus.	
Device index	Here is '1'	

The following is the command running result:

```
./spv_ipmi -I lanplus -H 192.168.9.172 -U root -P 0penBmc -C 17 isc mntiso umount 1

Success to unmount device 1
```

The following table is possible values:

Table 50-Possible values for running Unmount Web CD-ROM image

Possible Value
`Device index is invalid`
`Failed to allocate memory block`
`Success to unmount device 1`
`Failed to unmount device 1`
`Device 1 not inserted!`